

Hitesh Dammu

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MS Biomedical Engineering (ASU, 2023). Twenty months of clinical EEG work with three partner clinics: on-site data collection and a vendor-agnostic signal pipeline. Single-author methods preprint on what a fixed preprocessing contract restores for pretrained EEG classifiers under acquisition shift (v2, September 2026), with a public repository. Seeking doctoral training in human neurophysiology and closed-loop neuromodulation.

RESEARCH AND PROFESSIONAL EXPERIENCE

Signal pipeline engineer, clinical EEG software (ENCEPHLIAN) — Hyderabad, India — Jan 2025 to present

- Wrote readers for native Natus, Cadwell, Nihon Kohden, Persyst, Nicolet, BrainVision and EDF/BDF exports and a converter to one canonical representation (19-channel 10–20, 250 Hz, average reference, robust z-score).
- Collected EEG on-site at three partner clinics: acquisition setup, montage and electrode checks, session QC on legacy recorders; characterised each site's export format from its retrospective archive (weakly labelled, over 400 GB per clinic, vendor-locked).
- Converted the seven public Temple University Hospital EEG subcorpora (84,552 recordings) into the same representation for training and evaluation.
- Measured which acquisition differences the representation undoes for three frozen public models; single-author preprint with a public repository.
- Trained an interictal discharge detector on frozen EEGPT features (held-out AUC 0.929, $n = 476$) and a normal/abnormal triage model; built the EEG viewer and structured-report front end they feed into.
- Running an evaluation of the pipeline on the three clinics' retrospective and prospective recordings.
- Tools: Python (MNE, NumPy, SciPy, PyTorch, ONNX), TypeScript, PostgreSQL, Azure.

Research assistant, Neural Engineering Laboratory, Arizona State University — Jan 2022 to Apr 2023

- MATLAB time–frequency pipelines for invasive neural recordings; assisted non-human primate surgical preparation under IACUC protocol.
- Reconstructed and time-aligned an archival 32-channel micro-EEG recording from Wernicke's area in a locked-in patient, enabling downstream speech analysis.

Pilot study: delta-band micro-EEG dynamics during speech — with Barrow Neurological Institute — 2023

- Chronux spectral analysis of 32-channel micro-EEG from Wernicke's area and facial motor cortex; aligned phoneme boundaries to delta-band (1–4 Hz) phase; exploratory phoneme classification. Unpublished.

Behavioral health technician, Sevita Health — Phoenix, USA — Jun 2024 to Jan 2025

- Direct care for adults with serious mental illness and developmental disabilities; HIPAA-compliant documentation; interdisciplinary care teams.

PUBLICATIONS AND MANUSCRIPTS

- **Dammu H.** ESF: Harmonizing heterogeneous EEG acquisition for pretrained classifiers. Preprint v2, September 2026. [doi:10.5281/zenodo.22772008](https://doi.org/10.5281/zenodo.22772008). Single author; three frozen public models, five simulated acquisition shifts. Code: github.com/h5371h/esf-recoverability (MIT).
- Chary Nalband S, Pamidimukkala K, Akkapally P, Challa M, Kirupakaran Y, **Dammu H**, Thati A, Abraham S, Gunda R. Evaluation of surface roughness and cutting temperature of EN31 steel with varying MQL parameters. *IOP Conf. Ser.: Mater. Sci. Eng.* 895, 012003 (2020). [doi:10.1088/1757-899x/895/1/012003](https://doi.org/10.1088/1757-899x/895/1/012003). Undergraduate co-author, ICMM 2020, Tokyo.

EDUCATION

MS Biomedical Engineering, Arizona State University — Jan 2022 to Dec 2023

Relevant coursework (grade): Introduction to Neural Engineering (A), Data Analysis in Neuroscience (A), Mathematical Neuroscience (A), Magnetic Resonance Imaging (A–), Low-Power Bioelectronics (A+).

MS applied project (BME 593, grade A): in vivo and ex vivo murine benchtop electrophysiology under Daniel Gulick, BEST Lab, ASU.

B.Tech Mechatronics, Jawaharlal Nehru Technological University, Hyderabad — 2016 to 2021 (degree conferred April 2021). WES-verified US bachelor's equivalency: badges.wes.org

COURSEWORK AND INDEPENDENT PROJECTS IN NEURAL DATA ANALYSIS

Human single-neuron working memory analysis (independent, Sep 2026)

Rutishauser lab Sternberg working memory release (DANDI 000469; 902 units, 21 patients, 21 sessions, streamed from the archive in NWB). Reimplemented the lab's published selectivity tests in Python: 119 concept cells (13%, concentrated in amygdala and hippocampus) and 273 maintenance cells (30%, most frequent in pre-SMA and dACC). Added a memory-load analysis: accuracy fell and reaction time rose with load (Friedman $p < 0.005$), but maintenance-period firing rate did not scale with load at the population level, consistent with the lab's report that load is carried by a combined phase-rate code. Code, figures and write-up on portfolio. Python, PyNWB, SciPy.

Human MTL single-neuron memory selectivity (APM 598 / MAT 494, group project, instructor Dr. Sharon Crook, Spring 2023)

Rutishauser lab recognition memory dataset (DANDI 000004; 89 sessions, 59 patients). Adapted the lab's NWB pipeline; per-neuron rasters and PSTHs; classified visually selective vs memory selective cells by one-way ANOVA across image categories and bootstrap test on new vs old trials. Python, PyNWB.

Human dIPFC Neuropixels LFP and spiking (APM 598, individual, Spring 2023)

Paulk et al. 2022 intraoperative dataset (DANDI 000397, 384-channel probe). NWB extraction, raster plots, LFP spectrograms. Python, PyNWB.

Non-human primate array electrophysiology (BME 526, Spring 2022)

Spike sorting from 30 kHz Utah-array recordings: thresholding, PCA feature extraction, k-means clustering, SNR. Motor-cortex 96-electrode array during finger movements: event-aligned rasters, PETHs, finger-identity decoding from firing-rate modulation. LIP local field potentials: Chronux multitaper spectrograms across saccade targets and band-limited tuning curves. MATLAB.

Mathematical Neuroscience (APM 531) — analytical and numerical solutions of neural PDE models; neuron and dendritic dynamics.

SKILLS

- Signal processing: EEG/ECOG/sEEG preprocessing, re-referencing, multitaper and wavelet time-frequency analysis, phase estimation, spike sorting.
- Programming: Python (MNE, PyNWB, NumPy, SciPy, PyTorch), MATLAB (Chronux, FieldTrip), TypeScript, SQL.
- Data standards: NWB, BIDS, EDF/EDF+, vendor-native EEG formats.
- Ethics and protocols: CITI human-subjects research; IACUC (rodent and non-human primate).
- Self-study, verifiable: Probabilistic Graphical Models Specialization, Stanford Online (Mar 2020, [verify](#)); IBM Deep Learning Professional Certificate, edX (Mar 2020, [verify](#)); AI for Medicine Specialization, DeepLearning.AI (May 2021, [verify](#)); Deep Learning Specialization, DeepLearning.AI (May 2021, [verify](#)); IBM AI Engineering Professional Certificate ([badge](#)); Machine Learning, Stanford Online ([verify](#)); g.tec BCI and Neurotechnology Spring School 2022.
- Languages: English; willing to learn Dutch.

REFEREES

- Prof. Sharon Crook, School of Mathematical and Statistical Sciences, Arizona State University. Sharon.Crook@asu.edu
- Daniel Gulick, BioElectrical Systems and Technology Lab, Arizona State University (supervised MS applied project). Daniel.Gulick@asu.edu